# Quantitative Research

## Part 3: Statistical Inference

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# The lecture at a glance

| | Part | Content |
|---|---|---|
| | 1 – Descriptive Statistics | Basic concepts, levels of measurement, frequencies, measures of central tendency and dispersion |
| | 2 – Graphical Representations | Empirical CDF, histogram, boxplot |
| ▶ | **3 – Statistical Inference** | **Confidence intervals, hypothesis tests, $p$-values, one-sided tests, types of error** |
| | 4 – Regression | Correlation, simple and multiple linear regression, inference for regression, diagnostics |
| | 5 – Classification | Logistic regression, classification rules, confusion matrix, ROC/AUC, Bayes classifiers |

All analyses are carried out in **R** (with RStudio); every deck comes with an R script that reproduces its numbers.

# Contents – Part 3

1. Confidence Intervals
2. Hypothesis Tests
3. The $p$-Value
4. One-Sided Tests
5. Types of Error
6. Summary

# The common thread of this part

## One question runs through every section

We draw a **sample** and want to learn something about the **population** from it – while being *honest* about the remaining uncertainty.

**Two tools, one idea:**

- **Confidence interval** – "Which values of $\mu$ are *compatible* with the data?"
- **Hypothesis test** – "Is a specific value $\mu_0$ compatible – or do the data *reject* it?"

**One running story:**

### The dairy

A machine fills milk. Is it working correctly? We measure, estimate and test – and follow this example throughout the entire part.

# Learning objectives – Part 3

After this part you will be able to ...

- compute and correctly interpret a **confidence interval** for $\mu$,
- carry out a **hypothesis test** (known/unknown variance, one-/two-sided),
- compute the $p$-**value** and read it as a measure of evidence,
- distinguish **Type I and Type II errors** as well as the **power** of a test,
- read **critical values** from the $z$- and $t$-tables and link them to the $p$-value,
- implement every procedure in R.

**Key takeaway**

Common thread: infer from a sample – and report the uncertainty honestly.

# List of symbols (1/2)

| Symbol | Meaning | Symbol | Meaning |
|---|---|---|---|
| $\mu$ | Expected value | $n$ | Sample size |
| $\sigma^2$ | Variance | $X_i$ | Sample variable |
| $\sigma$ | Standard deviation | $x_i$ | Observed value |
| $\bar{X}$ | Sample mean | $\mu_0$ | Hypothesized value |
| $\bar{x}$ | Realization | $s^2$ | Sample variance |
| $s$ | Sample std. dev. | $H_0$ | Null hypothesis |
| $H_1$ | Alternative hypothesis | $\alpha$ | Type I error prob. |
| $\beta$ | Type II error prob. | $1-\beta$ | Power |

Convention: $N(\mu, \sigma^2)$ – the second argument is the **variance** (not the standard deviation).

# List of symbols (2/2)

| Symbol | Meaning | Symbol | Meaning |
|---|---|---|---|
| $Z_n$ | Test stat., known var. | $T_n$ | Test stat., unknown var. |
| $z_{1-\alpha/2}$ | Two-sided quantile, $N(0,1)$ | $t_{1-\alpha/2,n-1}$ | Two-sided quantile, $t_{n-1}$ |
| $z_{1-\alpha}$ | One-sided quantile, $N(0,1)$ | $t_{1-\alpha,n-1}$ | One-sided quantile, $t_{n-1}$ |
| $\Phi(\cdot)$ | CDF of $N(0,1)$ | $F_{t,n-1}$ | CDF of $t_{n-1}$ |
| $p$ | $p$-value | CI | Confidence interval |
| $\alpha$ | Significance level | $1-\alpha$ | Confidence level |
| $\beta$ | Type II error prob. | $1-\beta$ | Power |

# Overview

1 Confidence Intervals
- Motivation and intuition
- Known variance
- Length of the confidence interval
- Unknown variance
- Exercises

2 Hypothesis Tests

3 The $p$-Value

4 One-Sided Tests

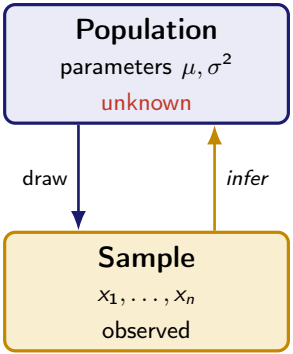
5 Types of Error

6 Summary

# Why inferential statistics?

## The starting question

We only know one **sample** – but want to say something about the **whole population**.




- **Descriptive:** describes the data at hand.
- **Inferential:** infers *back* from the sample to the population.
- This inference is *uncertain* – we must report the uncertainty.

# A single estimate is not enough

## Intuition

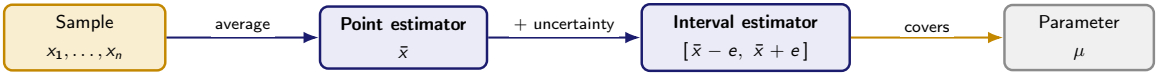
From the sample we compute the mean $\bar{x}$. But it almost **never** hits the true value $\mu$ exactly – a second sample would give a different value.

- Instead of a single point we report a **range** of values that are *compatible with the data*.
- The more precise our estimate, the *narrower* this range.

## Key takeaway

A **confidence interval** reports the estimate *together with its uncertainty* – a range of plausible values for $\mu$. (It is *not* a probability statement about where $\mu$ lies – see the interpretation slides.)

# From the sample to the interval

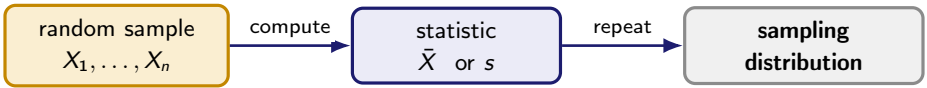



- The point estimator $\bar{x}$ is the *best single estimate*.
- The *margin of error* $e = z_{1-\alpha/2}\,\sigma/\sqrt{n}$ (half the interval's width) turns it into an interval – built so that the *procedure* covers the fixed true $\mu$ in $(1 - \alpha)$ of samples.

# Estimators are random variables

The sample $X_1, \ldots, X_n$ is **random**, so *anything* computed from it – $\bar{X}$, $s$, $Z_n$, $T_n$ – is itself a **random variable**.

- New sample $\Rightarrow$ a new $\bar{x}$ *and* a new $s$.
- Its distribution over *all possible samples* is the **sampling distribution** – so $\bar{X}$ and $s$ each have their own.




## Key takeaway

A confidence interval or test is built on the *sampling distribution* of an estimator – never on a single number alone.

# The standard error: how much $\bar{X}$ wobbles

## Definition

The **standard error** (SE) is the *standard deviation of the sampling distribution* of an estimator – the spread of $\bar{X}$ from sample to sample, *not* the spread of the raw data.

$$\mathrm{SE}(\bar{X}) = \frac{\sigma}{\sqrt{n}} \quad \Big(\text{or } \tfrac{s}{\sqrt{n}} \text{ when } \sigma \text{ is estimated}\Big).$$

- It is the natural **ruler** for inference: $Z_n$ and $T_n$ count *how many standard errors* $\bar{x}$ sits away from $\mu_0$.
- Larger $n \Rightarrow$ smaller SE $\Rightarrow$ narrower CI and sharper test.

## Key takeaway

Standard *deviation* = spread of the data; standard *error* = spread of the estimate.

# Why divide by the standard error?

## The raw deviation is not enough

A sample mean $\bar{x} = 8$ lies 2 units from a reference value $\mu_0 = 10$. Whether that gap is *small or large* depends entirely on the standard error – how much $\bar{X}$ wobbles.

| Scenario | $\mathrm{SE} = \sigma/\sqrt{n}$ | $z = \dfrac{8-10}{\mathrm{SE}}$ | reading |
|---|---|---|---|
| large spread, small $n$ $(\sigma = 6,\ n = 9)$ | 2.0 | $-1.0$ | ordinary |
| small spread, large $n$ $(\sigma = 4,\ n = 100)$ | 0.4 | $-5.0$ | far out |

## Key takeaway

Same raw deviation, opposite readings. Dividing by the SE turns it into a *universal* ruler – "how many standard errors" – comparable across samples and units.

# How we read probability: the frequentist view

## Fixed vs. random

- Parameters $\mu, \sigma$ are **fixed** (unknown) constants – not random.
- The data – and all we build from them ($\bar{X}$, $s$, the CI bounds, $Z_n$, $p$) – are **random**.

## What "probability" means here

The long-run *relative frequency* of an event over (hypothetically) very many repeated samples.

## Key takeaway

"95% confidence" and "5% significance" describe the **procedure across repeated samples** – not one particular interval and not one particular dataset.

# Our story: filling milk

## Example with known variance

- A dairy fills milk (label: $1\,\text{l} = 1000\,\text{ml}$).
- Machine with **known** variance $\sigma^2 = 9\,\text{ml}^2$, i.e. $\sigma = 3\,\text{ml}$.
- After servicing: is it still working correctly?
- Sample $n = 9$ (in ml):

1005, 1001, 997, 1000, 1001, 998, 999, 999, 1009

**Key figures**

$$\bar{x} = 1001\,\text{ml}$$

$$1 - \alpha = 0.95 \;\Rightarrow\; \alpha = 0.05$$

**Goal:** confidence interval for the true mean $\mu$.

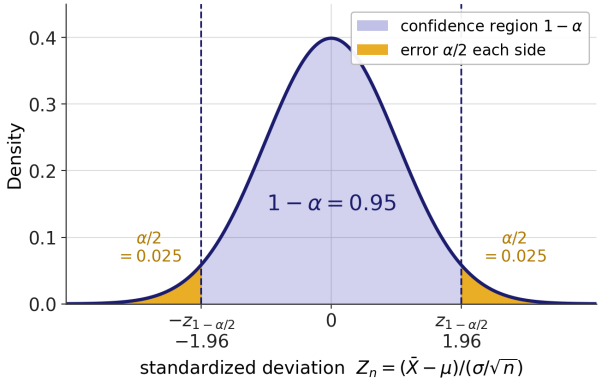
# How does the mean fluctuate? – (1) intuition

## Thought experiment

Imagine we draw the sample *over and over again*. Each time we get a slightly different $\bar{x}$.

- These $\bar{x}$-values scatter around the true value $\mu$.
- But they scatter **more tightly** than the individual values – the average evens out outliers.
- The standard error shrinks with $\sqrt{n}$: four times as much data $\Rightarrow$ half the spread *of the sample mean*.
- Reason: variances *add* for independent values, so the spread grows like $\sqrt{n}$ for the *sum* but shrinks like $1/\sqrt{n}$ once we *average*. (For non-normal data the **Central Limit Theorem** makes $\bar{X}$ approximately normal for large $n$.)

# How does the mean fluctuate? – (2) graphically




Distribution of the standardized statistic $Z_n \sim N(0,1)$: the central (blue) region holds 95 % of the probability – in 95 % of samples $Z_n$ lands here; 2.5 % in each tail.

# How does the mean fluctuate? – (3) formally

## Sampling distribution of the sample mean

For $X_i \sim N(\mu, \sigma^2)$ i.i.d., the *sampling distribution* of the random variable $\bar{X}$ is

$$\bar{X} \sim N\Big(\mu,\ \tfrac{\sigma^2}{n}\Big) \quad \Longrightarrow \quad Z_n := \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1).$$

Why $\sigma^2/n$? Averaging $n$ independent draws divides the variance by $n$ ($\mathrm{Var}(\bar{X}) = \sigma^2/n$), so the standard error is $\sigma/\sqrt{n}$; $Z_n$ is just $\bar{X}$ centred and scaled to $N(0,1)$.

## Two-sided probability statement

$$P\big(-z_{1-\alpha/2} \le Z_n \le z_{1-\alpha/2}\big) = 1 - \alpha$$

**In words:** With probability $1 - \alpha$, $Z_n$ lies between the two quantiles $\pm z_{1-\alpha/2}$.

# Derivation (1/2): rearrange step by step

Start from the pivot $Z_n = \dfrac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$, which lies within $\pm z_{1-\alpha/2}$ with probability $1-\alpha$:

$$1-\alpha = P\Big(-z_{1-\alpha/2} \;\le\; \tfrac{\bar{X}-\mu}{\sigma/\sqrt{n}} \;\le\; z_{1-\alpha/2}\Big) \qquad \triangleright \text{ starting point}$$

$$= P\Big(-z_{1-\alpha/2}\tfrac{\sigma}{\sqrt{n}} \;\le\; \bar{X} - \mu \;\le\; z_{1-\alpha/2}\tfrac{\sigma}{\sqrt{n}}\Big) \qquad \triangleright \text{ multiply all 3 parts by } \tfrac{\sigma}{\sqrt{n}} > 0$$

$$= P\Big(-\bar{X} - z_{1-\alpha/2}\tfrac{\sigma}{\sqrt{n}} \;\le\; -\mu \;\le\; -\bar{X} + z_{1-\alpha/2}\tfrac{\sigma}{\sqrt{n}}\Big) \qquad \triangleright \text{ subtract } \bar{X} \text{ everywhere}$$

**In words:** Multiplying by a *positive* number keeps the "$\le$" signs; we only used equivalence steps, so the probability stays exactly $1-\alpha$.

# Derivation (2/2): from $-\mu$ to the interval

$$1-\alpha = P\Big(\bar{X} + z_{1-\alpha/2}\tfrac{\sigma}{\sqrt{n}} \;\geq\; \mu \;\geq\; \bar{X} - z_{1-\alpha/2}\tfrac{\sigma}{\sqrt{n}}\Big) \qquad \triangleright \text{ multiply by } (-1): \text{ signs flip}$$

$$= P\Big(\bar{X} - z_{1-\alpha/2}\tfrac{\sigma}{\sqrt{n}} \;\leq\; \mu \;\leq\; \bar{X} + z_{1-\alpha/2}\tfrac{\sigma}{\sqrt{n}}\Big) \qquad \triangleright \text{ just read it left-to-right}$$

## Key takeaway

Confidence interval with **known** variance:

$$\mathrm{CI}_{1-\alpha} = \left[\bar{X} - z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}},\; \bar{X} + z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}\right]$$

Multiplying an inequality by $-1$ **reverses** "$\leq$" into "$\geq$" – that is the only subtle step. What is random are the *bounds* (they contain $\bar{X}$); $\mu$ itself is a fixed unknown.

# The formula in words

$$\text{CI} = \Big[ \underbrace{\bar{X}}_{\text{estimate}} \pm \underbrace{z_{1-\alpha/2}}_{\substack{\text{critical} \\ \text{value}}} \cdot \underbrace{\frac{\sigma}{\sqrt{n}}}_{\text{standard error}} \Big]$$

- **Centre** $\bar{X}$: our best estimate of $\mu$.
- **Standard error** $\sigma/\sqrt{n}$: how strongly $\bar{X}$ fluctuates.
- **Critical value** $z_{1-\alpha/2}$: how many standard errors we go to the left/right (1.96 at 95 %).

# Milk example: calculation

**Given:** $\bar{x} = 1001$, $\sigma = 3$, $n = 9$, $\alpha = 0.05$, $z_{0.975} = 1.96$.

$$\text{margin of error } e = z_{1-\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} = 1.96 \cdot \frac{3}{\sqrt{9}} = 1.96$$

$$\text{lower bound} = 1001 - 1.96 = 999.04$$

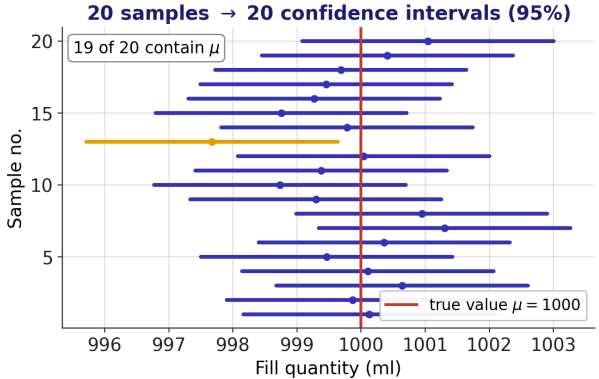
$$\text{upper bound} = 1001 + 1.96 = 1002.96$$

## Result

$$\text{CI}_{0.95} = \big[\, 999.04\,;\, 1002.96 \,\big] \text{ ml}$$

The hypothesized target value $\mu_0 = 1000$ lies *inside* the interval $\Rightarrow$ no evidence against a correctly working machine.

# What does "95 % confidence" mean?




Of 20 computed 95 % intervals, on average $\approx 19$ cover the true value $\mu$ (red line). The confidence *level* refers to the **procedure** across many samples.

# A common misconception

## Careful with the interpretation

"With 95 % probability $\mu$ lies in *this particular* interval" is, strictly speaking, **wrong**.

- $\mu$ is a fixed (unknown) number – it either is or is not inside the particular interval.
- What is random are the *bounds* of the interval, not $\mu$.
- Correct: "The **procedure** yields, in 95 % of samples, an interval that covers $\mu$."

# Three correct ways to read a confidence interval

1. **Coverage (the procedure).** The method produces intervals that contain the fixed true $\mu$ in a fraction $(1 - \alpha)$ of all samples.
2. **Compatibility.** The interval is the range of values for $\mu$ that are *consistent with the observed data*.
3. **Duality with the test.** It is exactly the set of null values $\mu_0$ that a two-sided level-$\alpha$ test would *not* reject.

**Key takeaway**

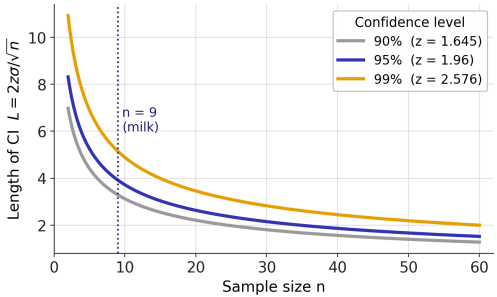
All three avoid the trap of putting a probability on *this one* interval: $\mu$ is fixed, the **interval** is what is random.

# What does the length depend on?

## Length

$$L = 2 \cdot z_{1-\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

- Higher confidence level $\Rightarrow$ larger $z_{1-\alpha/2}$ $\Rightarrow$ *longer*.
- Larger $n \Rightarrow$ *shorter* (with $1/\sqrt{n}$).
- Larger spread $\sigma \Rightarrow$ *longer*.




**In words:** More confidence *or* more precision – both require a larger sample.

# Different confidence levels – milk

| Confidence level | Quantile $z_{1-\alpha/2}$ | CI for milk example |
|---|---|---|
| 90 % | $z_{0.95} = 1.645$ | [999.36 ; 1002.65] |
| 95 % | $z_{0.975} = 1.96$ | [999.04 ; 1002.96] |
| 99 % | $z_{0.995} = 2.576$ | [998.42 ; 1003.58] |

**Key takeaway**

Higher confidence level $\Rightarrow$ wider interval $\Rightarrow$ less precise statement. Confidence and precision are a trade-off.

# More realistic: the variance is unknown

## Problem

In practice $\sigma^2$ is almost never known. We must *also estimate* it from the sample.

## Estimator of the variance

$$s^2 = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \bar{X})^2$$

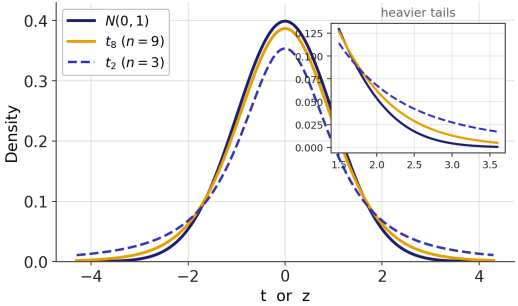
- We replace $\sigma$ by $s$ in the test statistic.
- This costs *accuracy* – and changes the distribution (see next slide).

# Why the $t$-distribution? – intuitively and graphically

## Standardized statistic (normal data)

$$T_n = \frac{\bar{X} - \mu}{s/\sqrt{n}} \sim t_{n-1}$$

- $s$ is **itself a random variable**.
- $T_n$ is a ratio of *two* random quantities $\Rightarrow$ **heavier tails**.
- For large $n$, $s$ barely varies, so $t_{n-1} \to N(0,1)$.




## Key takeaway

With unknown $\sigma$ always use the $t_{n-1}$-distribution – its quantiles are larger, so the CI is wider.

# Milk example with unknown variance

## Confidence interval

$$\mathrm{CI} = \left[\bar{X} - t_{1-\alpha/2,\, n-1}\frac{s}{\sqrt{n}},\ \bar{X} + t_{1-\alpha/2,\, n-1}\frac{s}{\sqrt{n}}\right]$$

**Data:** $\bar{x} = 1001$, $n = 9$, $\alpha = 0.05$, $s = 3.7749$, $t_{0.975;\,8} = 2.3060$.

$$e = 2.3060 \cdot \frac{3.7749}{\sqrt{9}} = 2.9017$$

$$\mathrm{CI}_{0.95} = [\,1001 \pm 2.9017\,] = [\,998.10\,;\,1003.90\,]$$

## Comparison

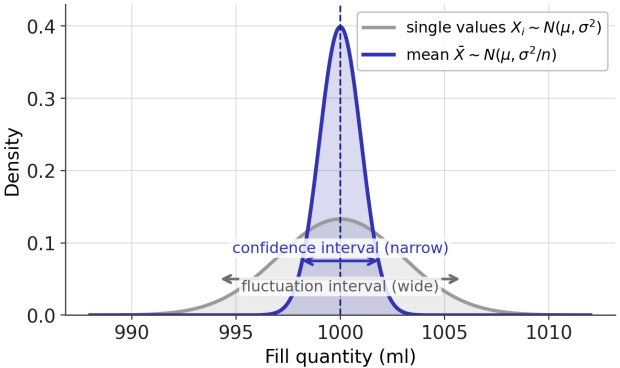
Wider than [999.04 ; 1002.96] (known variance) – the price for estimating $\sigma$.

# Summary: known vs. unknown variance

| | **known $\sigma$** | **unknown $\sigma$** |
|---|---|---|
| Spread used | $\sigma$ | $s$ (estimated) |
| Distribution | $N(0,1)$ | $t_{n-1}$ |
| Quantile (95 %, $n = 9$) | 1.96 | 2.31 |
| CI | narrower | wider |

- For large $n$ ($n \gtrsim 30$) the difference is small ($t_{n-1} \approx N(0,1)$).
- For small samples it is clearly noticeable.

# Do not confuse: CI vs. fluctuation interval




- **Fluctuation interval:** spread of *individual* values $X_i$ around $\mu$ – uses $\sigma$ (wide).
- **Confidence interval:** uncertainty about $\mu$ – uses $\sigma/\sqrt{n}$ (narrow).

**In words:** Both intervals share the same centre but answer completely different questions.

# Exercise – filling cream

## Problem statement

A dairy fills cream into cups (label: 250 g). The variance of the machine is known: $\sigma^2 = 4\,\text{g}^2$. After servicing, $n = 16$ cups are sampled (in g):

$$251, 248, 248, 249, 250, 251, 247, 246, 252, 251, 248, 247, 246, 249, 248, 252$$

Confidence level $1 - \alpha = 0.95$.

**Required:** confidence interval for $\mu$ – (a) with known, (b) with unknown variance.

# Solution (1/2) – cream, known variance

**Rule:** $\mathrm{CI} = \left[\bar{x} \pm z_{1-\alpha/2}\,\sigma/\sqrt{n}\right]$.
**Intermediate values:** $\bar{x} = 248.9375$, $\sigma = 2$, $n = 16$, $z_{0.975} = 1.96$.

$$e = 1.96 \cdot \frac{2}{\sqrt{16}} = 1.96 \cdot 0.5 = 0.98$$
$$\mathrm{CI}_{0.95} = [\,248.9375 \pm 0.98\,] = [\,247.96\,;\,249.92\,]$$

## Interpretation

The target value $\mu_0 = 250$ lies *outside* the CI $\Rightarrow$ with known variance: indication that the machine is **not** working correctly.

# Solution (2/2) – cream, unknown variance

**Rule:** as above, but $z \to t_{1-\alpha/2,\,n-1}$ and $\sigma \to s$.
**Intermediate values:** $s^2 = 4.0625 \Rightarrow s = 2.0156,\ t_{0.975;\,15} = 2.1314.$

$$e = 2.1314 \cdot \frac{2.0156}{\sqrt{16}} = 1.0740$$
$$\mathrm{CI}_{0.95} = [\,248.9375 \pm 1.0740\,] = [\,247.86\,;\,250.01\,]$$

## Comparison

Now $\mu_0 = 250$ lies (just) *inside* the CI – the verdict turns out *differently* than with known variance. The wider $t$-interval is more cautious.

# Summary and outlook

- Confidence intervals quantify the **uncertainty** of a point estimate.
- Known variance: quantiles of $N(0,1)$; unknown: of $t_{n-1}$.
- **Bridge to the test:** if the CI contains the hypothesized value $\mu_0$, there is no evidence against $H_0 : \mu = \mu_0$ – otherwise there is.

## Next section

We formalize this procedure as a **hypothesis test**.

# Overview

1 Confidence Intervals

2 Hypothesis Tests
- Idea and hypotheses
- From CI to test
- Known variance
- Unknown variance
- Exercises
- Tests with R

3 The $p$-Value

4 One-Sided Tests

5 Types of Error

6 Summary

# What is a hypothesis test? – intuition

## The basic idea, as in court

An assumption (the *null hypothesis* $H_0$) is held true until the data **speak strongly enough against it**.

- $H_0$ = "innocent" (status quo).
- We reject $H_0$ only on *convincing* evidence.
- Little evidence $\Rightarrow$ acquittal for lack of evidence – not "proven innocent".

## Milk

$H_0 : \mu = 1000$ – "the machine works correctly". Only a clear deviation from 1000 makes us doubt this.

# The two hypotheses

## Structure

$$H_0 : \mu = \mu_0 \qquad \text{vs.} \qquad H_1 : \mu \neq \mu_0$$

- We **always** argue under "$H_0$ is true".
- Why unequal? $H_0 : \mu = \mu_0$ fixes a *single* value, so the test statistic has *one* definite distribution we can compute with; $H_1$ is a whole *range*, with no single distribution.
- Two outcomes: *keep* $H_0$ or *reject* it.

## Key takeaway

What you want to **establish** statistically goes into $H_1$.

# Which hypothesis goes where?

## Rule of thumb

The statement you want to *find evidence for* – the "more important" claim – is formulated as $H_1$, because only the Type I error (wrongly rejecting $H_0$) can be controlled via $\alpha$.

| **You want evidence that ...** | $H_0$ | $H_1$ |
|---|---|---|
| the supplier's fill weight *deviates* from the spec | $\mu = \mu_0$ | $\mu \neq \mu_0$ |
| the mean processing time *changed* after the update | $\mu = \mu_0$ | $\mu \neq \mu_0$ |
| the mean daily revenue *differs* from the plan figure | $\mu = \mu_0$ | $\mu \neq \mu_0$ |

# Bridge: confidence interval and test are the same

## Testing with the CI

If $\mu_0$ lies *inside* the confidence interval, $H_0 : \mu = \mu_0$ is **not** rejected – if it lies outside, it is.

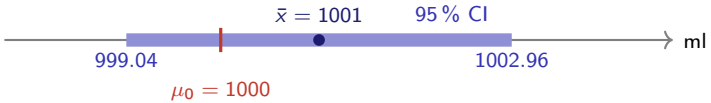
## Equivalence – same algebra, two readings

"$\mu_0$ inside the CI" means $|\bar{x} - \mu_0| \leq z_{1-\alpha/2}\,\sigma/\sqrt{n}$, which rearranges to $|Z_n| \leq z_{1-\alpha/2}$ – exactly the non-rejection condition. So a test (level $\alpha$) and a CI (level $1-\alpha$) **always** agree.

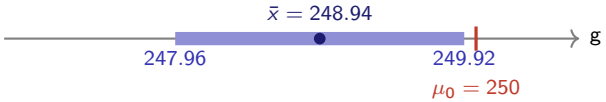
**In words:** The CI shows *all* $\mu_0$ that you would not reject.

# The duality on the number line

**Milk:** $\mu_0 = 1000$ lies *inside* the CI $\Rightarrow$ **keep** $H_0$



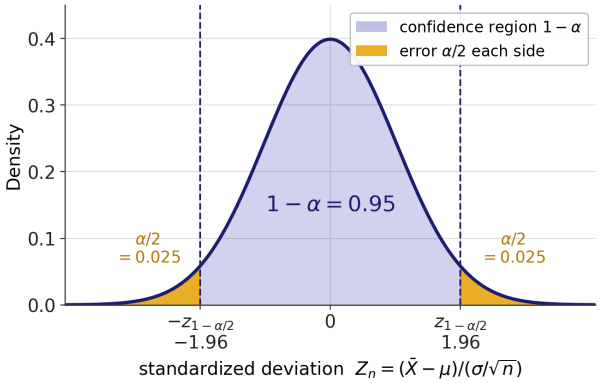

**Cream:** $\mu_0 = 250$ lies *outside* the CI $\Rightarrow$ **reject** $H_0$




## Key takeaway

$\mu_0$ inside the CI $\Leftrightarrow$ keep $H_0$; outside $\Leftrightarrow$ reject. (Both at known $\sigma$, $\alpha = 0.05$.)

# One picture, two readings




- **Distribution reading:** the central blue region holds 95 % of the $Z_n$ distribution (in 95 % of samples $Z_n$ lands here).
- **Test reading:** blue = **acceptance region** (= non-rejection region), orange = **rejection region** ($\alpha$).
- Test statistic in the orange tail $\Rightarrow$ reject $H_0$.

"Acceptance region" is conventional for the *non-rejection* region: landing there means we *fail to reject* $H_0$, not that we accept it as true.

# The critical value – the bar you must clear

## Plain idea

The **critical value** is a fixed threshold on the test statistic. It splits all possible outcomes into two zones:

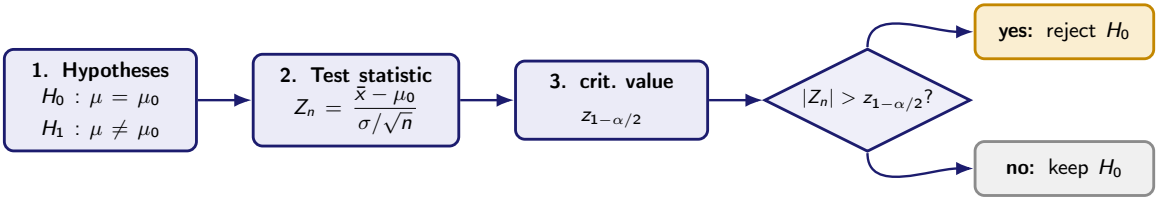
### Keep $H_0$

Statistic *inside* the threshold – the deviation is small enough to pass as ordinary chance.

### Reject $H_0$

Statistic *beyond* the threshold – too extreme to be chance at level $\alpha$.

# The test recipe (known variance)




- **Key move:** plug the hypothesized $\mu_0$ in for the unknown $\mu$ – *if* $H_0$ holds then $Z_n \sim N(0,1)$, so a $|Z_n|$ that would be very unusual for $N(0,1)$ is evidence against $H_0$.
- $|Z_n| \leq z_{1-\alpha/2}$: acceptance (non-rejection) region; beyond it: rejection region (area $\alpha$ under $H_0$). Fix $\alpha$ *in advance*.

# Milk test with known variance

**Data:** $\bar{x} = 1001$, $\sigma = 3$, $n = 9$, $\mu_0 = 1000$, $\alpha = 0.05$, $z_{0.975} = 1.96$.

$$Z_n = \frac{1001 - 1000}{3/\sqrt{9}} = \frac{1}{1} = 1.000$$

$$|Z_n| = 1.000 \; < \; 1.96 = z_{0.975}$$

## Decision

$H_0 : \mu = 1000$ is **not** rejected – no evidence against a correctly working machine.

**In words:** Same result as the CI: $\mu_0 = 1000 \in [999.04\,;\,1002.96]$.

# Test with unknown variance

## Only two changes to the recipe

$$T_n = \frac{\bar{X} - \mu_0}{s/\sqrt{n}} \sim t_{n-1} \text{ under } H_0, \qquad \text{rejection: } |T_n| > t_{1-\alpha/2,\, n-1}.$$

- $\sigma \to s$ (estimated) and $z \to t_{n-1}$.
- **Assumption:** data (approximately) normally distributed *or* $n \geq 30$ (central limit theorem).

# Milk test with unknown variance

**Data:** $\bar{x} = 1001$, $s = 3.7749$, $n = 9$, $\mu_0 = 1000$, $\alpha = 0.05$, $t_{0.975;\,8} = 2.3060$.

$$T_n = \frac{1001 - 1000}{3.7749/\sqrt{9}} = \frac{1}{1.2583} = 0.7947$$

$$|T_n| = 0.7947 \;<\; 2.3060$$

## Decision

$H_0 : \mu = 1000$ is **not** rejected. (More cautious $t$-quantile 2.31 instead of 1.96 – the result is nevertheless the same here.)

# Before you test – the assumptions checklist

1. **Random sample, independent observations** – the $X_i$ are i.i.d. draws from the population. (No clustering, no repeated measurements of the same unit.)
2. **Distribution:** data approximately normal *or* $n \gtrsim 30$ – then the CLT makes $\bar{X}$ approximately normal.
3. **Spread:** $\sigma$ known $\rightarrow$ $z$-test; $\sigma$ estimated by $s \rightarrow$ $t$-test with $\mathrm{df} = n - 1$.
4. **Question fixed in advance:** $\mu_0$, the direction (one-/two-sided) and $\alpha$ are chosen *before* seeing the results.

## Key takeaway

If 1.–2. fail, the stated $\alpha$ is not honest: the 5 % false-alarm guarantee only holds under the model assumptions.

# Exercise – cream test

## Problem statement

Cream, label 250 g, known variance $\sigma^2 = 4$. Sample $n = 16$, $\bar{x} = 248.9375$. Significance level $\alpha = 0.05$, $\mu_0 = 250$.

Carry out the test – (a) known variance, (b) unknown variance ($s = 2.0156$, $t_{0.975;\,15} = 2.1314$).

# Solution (1/2) – cream, known variance

**Rule:** reject if $|Z_n| > z_{1-\alpha/2}$.

$$Z_n = \frac{248.9375 - 250}{2/\sqrt{16}} = \frac{-1.0625}{0.5} = -2.125$$

$$|Z_n| = 2.125 \; > \; 1.96 = z_{0.975}$$

## Decision

$H_0 : \mu = 250$ is **rejected**. There is evidence that the machine does not fill correctly.
(Consistent with the CI: $250 \notin [247.96\,;\,249.92]$.)

# Solution (2/2) – cream, unknown variance

$$T_n = \frac{248.9375 - 250}{2.0156/\sqrt{16}} = \frac{-1.0625}{0.5039} = -2.1086$$

$$|T_n| = 2.1086 \; < \; 2.1314 = t_{0.975;\,15}$$

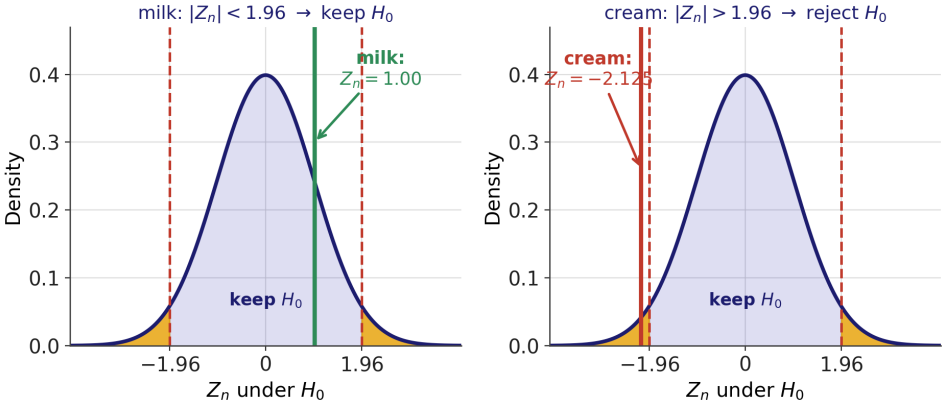
## Decision

$H_0 : \mu = 250$ is **not** rejected (just barely!).

## Key takeaway

Same data set, a different assumption about $\sigma$ – a different decision. The $t$-distribution is more conservative (heavier tails).

# Milk vs. cream – the picture behind both decisions




Two-sided test, known $\sigma$, $\alpha = 0.05$: the milk statistic ($Z_n = 1.00$) stays *inside* the bars $\pm 1.96$, the cream statistic ($Z_n = -2.125$) lands *beyond* them. The decision is nothing more than: **where does my statistic fall?**

# Exercise – reaction time (large sample)

## Problem

A training is meant to change the baseline reaction time $\mu_0 = 250\,\text{ms}$. A sample of $n = 50$ gives $\bar{x} = 242$, $s = 30$. Test *two-sided* at $\alpha = 0.05$.

**Hint:** $t_{0.975;\,49} = 2.0096$ (already close to 1.96).

# Solution – reaction time

$$T_n = \frac{242 - 250}{30/\sqrt{50}} = \frac{-8}{4.2426} = -1.8856, \qquad |T_n| = 1.8856 < 2.0096$$

## Decision & outlook

Keep $H_0$ (two-sided): $p = 2\,F_{t,49}(-1.8856) = 0.0653 > 0.05$. With $n = 50 \geq 30$ the CLT justifies the $t$-test. *But* the question may be directional – a one-sided test can decide differently (see the one-sided exercises).

# Hypothesis test with R – `t.test`

## Command (one-sample $t$-test)

```
milk ← c(1005, 1001, 997, 1000, 1001, 998, 999, 999, 1009)
t.test(milk, mu = 1000)
```

By default `t.test` performs a **two-sided** $t$-test with **unknown** variance and directly returns $t$, df, $p$-value and CI.

# Recap – and one limitation of the recipe

## The recipe in one line

Standardize the deviation, compare with the critical value, decide.

| Example | Statistic | Bar | Decision | How close? |
|---|---|---|---|---|
| milk, $\sigma$ known | $\lvert Z_n \rvert = 1.000$ | 1.960 | keep $H_0$ | clear call |
| cream, $\sigma$ known | $\lvert Z_n \rvert = 2.125$ | 1.960 | reject $H_0$ | close call |
| cream, $\sigma$ unknown | $\lvert T_n \rvert = 2.109$ | 2.131 | keep $H_0$ | razor-thin |

## Next section

All three verdicts are a bare *yes/no* – the last column is invisible in them. The $p$-**value** puts exactly this – the *strength* of the evidence – on a continuous 0–1 scale.

# Overview

1 Confidence Intervals

2 Hypothesis Tests

3 The *p*-Value
- Concept
- Calculation
- Exercises

4 One-Sided Tests

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# The $p$-value – intuition

## A single question

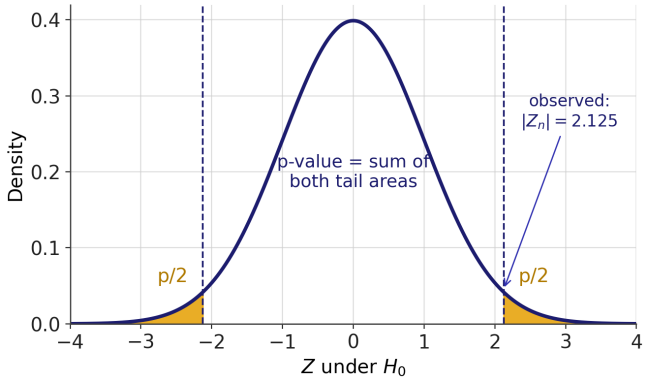
"How **surprising** is my data if $H_0$ were true?"

- **Small $p$-value** ($\approx 0$): such an extreme result would be very rare under $H_0 \Rightarrow$ doubt about $H_0$.
- **Large $p$-value** ($\approx 1$): the data are unsurprising under $H_0 \Rightarrow$ no reason to doubt it (but *not* positive proof of $H_0$).

## Key takeaway

The $p$-value measures *how surprising* the data are if $H_0$ holds – the smaller, the stronger the evidence **against** $H_0$.

# The *p*-value – graphically




The (two-sided) *p*-value is the **total area** in both tails that is *at least as extreme* as the observed value.

# The *p*-value – definition

## Definition

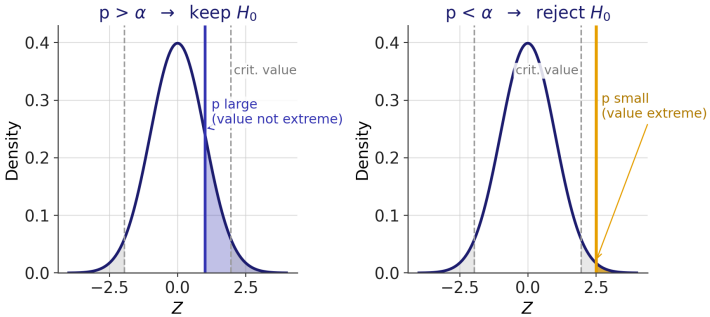
The *p*-value is the probability, under the validity of $H_0$, of observing a test statistic that is **at least as extreme** as the one actually observed.

- **"At least as extreme" is set by $H_1$:** two-sided ($\mu \neq \mu_0$) = both tails $|Z| \geq |Z_n|$; one-sided = only the tail in the $H_1$ direction. Same statistic, different $p$.
- $0 \leq p \leq 1$; software (e.g. `R`) always reports it.
- **Advantage:** more informative than a bare yes/no – it quantifies the *strength* of the evidence.

# Decision rule

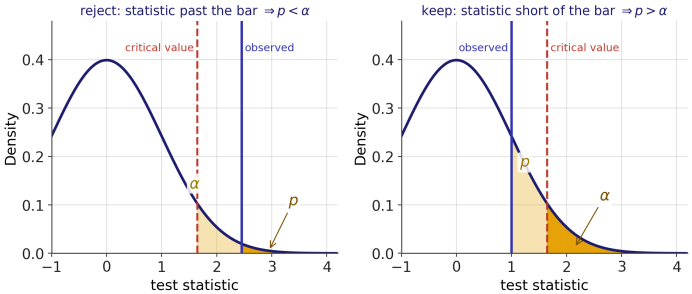
## One rule for all tests

$$p < \alpha \implies \text{reject } H_0 \qquad p \geq \alpha \implies \text{keep } H_0$$




- Left: $p$ large, value in the acceptance region.
- Right: $p$ small, value in the rejection region.
- $\alpha$ is the "benchmark".

# Two rules, one decision: critical value ↔ *p*-value




- $\alpha$ = tail area beyond the **critical value** (fixed in advance).
- $p$ = tail area beyond your **observed** statistic.
- Statistic past the bar ⇔ its tail is smaller ⇔ $p < \alpha$.

## Key takeaway

The two decision rules are **identical**: "test statistic beyond the critical value" and "$p < \alpha$" always agree – the critical value is simply the statistic whose tail area equals $\alpha$.

# Both rules, one verdict – worked example

**Cream, two-sided, known $\sigma$:** $Z_n = -2.125$, $\alpha = 0.05$.

## Critical-value rule

$$|Z_n| = 2.125 > 1.96 = z_{0.975}$$

$$\Rightarrow \textbf{reject } H_0.$$

## *p*-value rule

$$p = 2\,\Phi(-2.125) = 0.034$$

$$0.034 < 0.05 \Rightarrow \textbf{reject } H_0.$$

## Key takeaway

Same data, same $\alpha$, **same decision** – because $|Z_n| > z_{0.975}$ and $p < \alpha$ are just two ways of saying the observed value sits past the bar.

# Computing the *p*-value (two-sided)

## Known variance

$$p = 2 \cdot \Phi\big(-|Z_n|\big) = P\big(|Z| > |Z_n|\big), \quad Z \sim N(0,1)$$

## Unknown variance

$$p = 2 \cdot F_{t,n-1}\big(-|T_n|\big)$$

**In words: Why double?** With $H_1 : \mu \neq \mu_0$ a deviation in *either* direction counts, so "at least as extreme" means *both* tails $|Z| > |Z_n|$; by symmetry each has area $\Phi(-|Z_n|)$ – hence the factor 2. (One-sided: a single tail, no doubling.)

# $p$-value – milk (two variants)

**known variance, $Z_n = 1.000$**

$$p = 2 \cdot \Phi(-1.00) = 2 \cdot 0.15866 = 0.3173$$

**unknown variance, $T_n = 0.7947$**

$$p = 2 \cdot F_{t,8}(-0.7947) = 2 \cdot 0.2249 = 0.4497$$

**Decision**

Both $p$-values $> 0.05 \Rightarrow H_0 : \mu = 1000$ not rejected. The value 0.4497 matches the `t.test` output.

# $p$-value – cream (two variants)

## known variance, $Z_n = -2.125$

$$\begin{aligned} p &= 2 \cdot \Phi(-2.125) \\ &= 2 \cdot 0.0168 = 0.0336 \end{aligned}$$

$0.0336 < 0.05 \Rightarrow$ reject.

## unknown variance, $T_n = -2.1086$

$$\begin{aligned} p &= 2 \cdot F_{t,15}(-2.1086) \\ &= 2 \cdot 0.0261 = 0.0522 \end{aligned}$$

$0.0522 > 0.05 \Rightarrow$ keep.

## Key takeaway

The same data set can be just *significant* or just *not significant* depending on the assumption about $\sigma$.

# *p*-value – typical misconceptions

## What the *p*-value is *not*

- **Not** the probability that $H_0$ is true.
- A large *p*-value does **not prove** $H_0$ – it only shows *no contradiction*.

## Correct – mind the direction

The *p*-value is the probability *of the data (or more extreme)* **assuming** $H_0$ (chance) holds: $P(\text{data} \mid H_0)$. It *conditions on* chance; it does **not** give the probability that chance was the *cause*, $P(H_0 \mid \text{data})$.

# Significant $\neq$ important – the large-$n$ trap

## The $p$-value measures evidence, not size

With enough data, *any* tiny deviation from $\mu_0$ becomes statistically significant – even one that is practically irrelevant.

## Milk machine, observed $\bar{x} = 1000.2$ ml ($\mu_0 = 1000$, $\sigma = 3$)

| | $Z_n$ | $p$ (two-sided) | verdict at $\alpha = 0.05$ |
|---|---|---|---|
| $n = 9$ | 0.20 | 0.84 | not significant |
| $n = 10\,000$ | 6.67 | $< 0.0001$ | **highly** significant |

## Key takeaway

The same irrelevant 0.2 ml deviation – invisible at $n = 9$, "highly significant" at $n = 10\,000$. So report the **effect size and the CI**, not just $p$.

# Exercise – $p$-value and the choice of $\alpha$

## Problem

A two-sided test with known variance yields $Z_n = 1.85$. Compute the $p$-value and decide at $\alpha = 0.05$ and at $\alpha = 0.10$.

**Hint:** $\Phi(1.85) = 0.9678$.

# Solution – $p$-value and $\alpha$

$$p = 2\left(1 - \Phi(1.85)\right) = 2 \cdot 0.0322 = 0.0643$$

- At $\alpha = 0.05$: $0.0643 > 0.05 \Rightarrow$ **keep** $H_0$.
- At $\alpha = 0.10$: $0.0643 < 0.10 \Rightarrow$ **reject** $H_0$.

**Key takeaway**

The same data can be "significant" or not – $\alpha$ must be fixed *in advance*, never chosen to fit the result.

# Exercise – interpret the $p$-value (concept)

## Which statements are correct? ($p = 0.03$, $\alpha = 0.05$)

1. "There is a 3 % probability that $H_0$ is true."
2. "If $H_0$ were true, data this extreme occur in 3 % of samples."
3. "We reject $H_0$ at $\alpha = 0.05$."
4. "The effect is large and important."

# Solution – interpret the $p$-value

## Answers

1. **Wrong**: $p$ is $P(\text{data} \mid H_0)$, not $P(H_0 \mid \text{data})$.
2. **Correct**: that is exactly the definition.
3. **Correct**: $0.03 < 0.05$.
4. **Wrong**: significance $\neq$ effect size; a tiny effect can be significant with large $n$.

## Key takeaway

A small $p$-value signals *evidence against* $H_0$ – not the probability of $H_0$, and not the size or importance of the effect.

# Overview

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2 Hypothesis Tests

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- Motivation
- Left- and right-sided
- One-sided $p$-value
- One-sided confidence intervals
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5 Types of Error

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# When only one direction matters

## The idea

In a two-sided test *any* departure from $\mu_0$ counts against $H_0$. Often, though, a deviation is a **problem in only one direction** – the other direction is harmless or even welcome.

## Too little in the bottle

For consumer protection only *underfilling* ($\mu < 1000$) is critical – overfilling does not harm the customer.

## Unemployment insurance

Only if the rate *exceeds* a threshold do contributions fall short. A lower rate causes no problem.

## Key takeaway

The *question* – not the data – decides the direction. The deviation we truly care about defines $H_1$, and the *whole* $\alpha$ goes on that one side instead of $\alpha/2$ on each.

# Business examples – one direction only

| You want evidence that ... | $H_1$ | Direction |
|---|---|---|
| the new ad campaign *raises* the conversion rate | $\mu > \mu_0$ | right-sided |
| the new process *lowers* the average unit cost | $\mu < \mu_0$ | left-sided |

In both cases $H_0 : \mu = \mu_0$, as before.

**Key takeaway**

As in the two-sided case, the claim goes into $H_1$ – but now its *inequality* also fixes the side on which we reject.

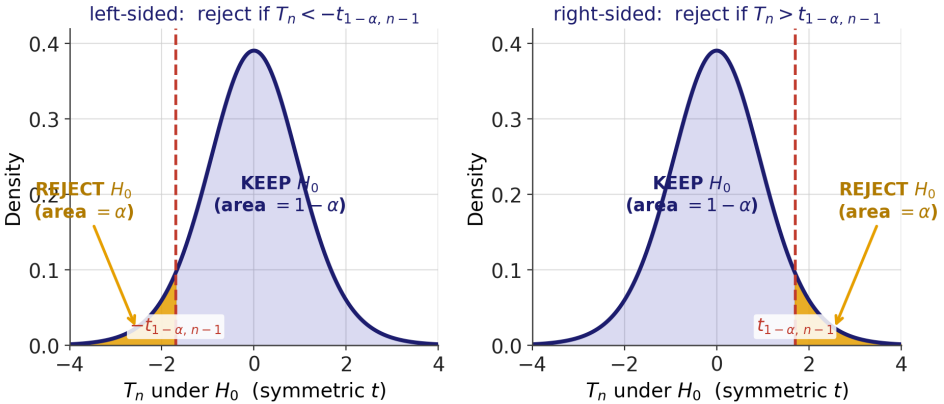
# How to find the critical value – 3 steps

1. **Direction** (read from $H_1$): one-sided or two-sided?
2. **Tail probability:**
   - two-sided $\rightarrow$ *split* $\alpha$: use the $1 - \alpha/2$ quantile ($\pm$ value),
   - one-sided $\rightarrow$ keep $\alpha$ *whole*: use the $1 - \alpha$ quantile (one value, on the $H_1$ side).
3. **Which table?** $\sigma$ known $\rightarrow$ $z$ (normal); $\sigma$ unknown $\rightarrow$ $t$ with $\mathrm{df} = n - 1$.

## $\alpha = 0.05$ – read the numbers off the table

known $\sigma$: two-sided $z_{0.975} = 1.96$, one-sided $z_{0.95} = 1.645$.
unknown $\sigma$, $n = 12$: two-sided $t_{0.975;11} = 2.20$, one-sided $t_{0.95;11} = 1.80$.

# One-sided critical values – left and right

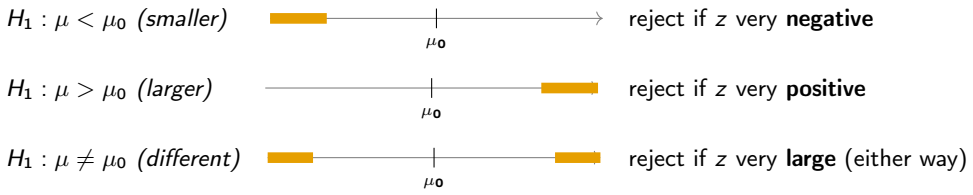



The whole rejection area $\alpha$ sits in *one* tail – left for "too small", right for "too large". The curve is the **symmetric** $t$-distribution: only the side changes, the shape never does.

# The golden rule of one-sided tests

## If you remember only one sentence

We reject $H_0$ when the sample result is **sufficiently extreme in the direction specified by the alternative hypothesis** $H_1$.




## Key takeaway

The inequality in $H_1$ points straight at the rejection region (orange): "$<$" $\rightarrow$ left tail, "$>$" $\rightarrow$ right tail, "$\neq$" $\rightarrow$ both.

# A left-sided test as a lawyer's argument

**Your claim** ($H_1 : \mu < 10$): "the true mean is *smaller* than 10." The sample mean $\bar{x}$ is your evidence:

| $\bar{x} = 8$ | $\bar{x} = 9$ | $\bar{x} = 10$ | $\bar{x} = 11$ | $\bar{x} = 12$ |
|---|---|---|---|---|
| strong *for* | some *for* | neutral | against | against |

- Below 10 $\Rightarrow$ evidence *for* $H_1$; if extreme enough, **reject** $H_0$.
- $\bar{x} = 12$ is far from 10 but in the *wrong* direction – it gives **no** support for "$\mu < 10$".

**Key takeaway**

The test asks whether the evidence *in the* $H_1$ *direction* is strong enough to reject $H_0$ – a big deviation the wrong way counts for nothing.

# Critical values of the standard normal $N(0,1)$

| Confidence $1-\alpha$ | $\alpha$ | **one-sided** $z_{1-\alpha}$ | **two-sided** $z_{1-\alpha/2}$ |
|---|---|---|---|
| 90 % | 0.10 | 1.282 | 1.645 |
| 95 % | 0.05 | 1.645 | 1.960 |
| 98 % | 0.02 | 2.054 | 2.326 |
| 99 % | 0.01 | 2.326 | 2.576 |
| 99.9 % | 0.001 | 3.090 | 3.291 |

**In words:** Pattern: the two-sided value at level $\alpha$ equals the one-sided value at level $\alpha/2$ – e.g. two-sided 95% = 1.96 = one-sided 97.5%.

# Critical values of the $t$-distribution $t_{\mathrm{df}}$

| | **upper-tail probability** (one-sided $\alpha$ \| two-sided $\alpha/2$) | | | | |
|---|---|---|---|---|---|
| $\mathrm{df} = n-1$ | 0.10 | 0.05 | 0.025 | 0.01 | 0.005 |
| 1 | 3.078 | 6.314 | 12.706 | 31.821 | 63.657 |
| 2 | 1.886 | 2.920 | 4.303 | 6.965 | 9.925 |
| 3 | 1.638 | 2.353 | 3.182 | 4.541 | 5.841 |
| 4 | 1.533 | 2.132 | 2.776 | 3.747 | 4.604 |
| 5 | 1.476 | 2.015 | 2.571 | 3.365 | 4.032 |
| 6 | 1.440 | 1.943 | 2.447 | 3.143 | 3.707 |
| 7 | 1.415 | 1.895 | 2.365 | 2.998 | 3.499 |
| 8 | 1.397 | 1.860 | 2.306 | 2.896 | 3.355 |
| 9 | 1.383 | 1.833 | 2.262 | 2.821 | 3.250 |
| 10 | 1.372 | 1.812 | 2.228 | 2.764 | 3.169 |
| 15 | 1.341 | 1.753 | 2.131 | 2.602 | 2.947 |
| 20 | 1.325 | 1.725 | 2.086 | 2.528 | 2.845 |
| 30 | 1.310 | 1.697 | 2.042 | 2.457 | 2.750 |
| 60 | 1.296 | 1.671 | 2.000 | 2.390 | 2.660 |
| $\infty$ | 1.282 | 1.645 | 1.960 | 2.326 | 2.576 |

Highlighted: $t_{0.975;\,9} = 2.262$ (two-sided 95%, $n = 10$); the $\infty$-row is the normal.

# How to read the tables – worked lookups

## Three questions, every time

(1) one- or two-sided? (2) which $\alpha$? (3) $z$ or $t$ (df $= n - 1$)?

**Normal** ($\sigma$ known):

- two-sided, $\alpha = 0.05 \to$ column $1 - \frac{\alpha}{2} = 0.975$ $\to z = 1.96$.
- one-sided, $\alpha = 0.05 \to 1 - \alpha = 0.95$ $\to z = 1.645$.

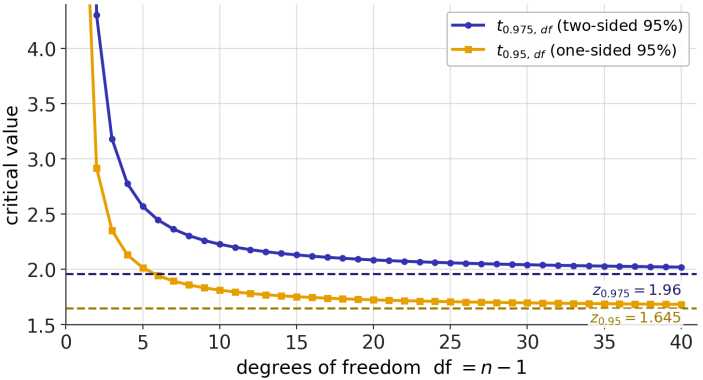
$t$ ($\sigma$ unknown), $n = 10 \Rightarrow$ df$= 9$:

- two-sided, $\alpha = 0.05 \to$ upper-tail 0.025 $\to t = 2.262$.
- one-sided, $\alpha = 0.05 \to$ upper-tail 0.05 $\to t = 1.833$.

## Key takeaway

Two-sided *halves* the tail probability ($\alpha \to \alpha/2$); the $t$-row is $\mathrm{df} = n - 1$. As df grows, the $t$-value slides toward $z$.

# Why the $t$-table ends in the normal




- Small df $\rightarrow$ heavier tails $\rightarrow$ *larger* critical value.
- As df grows, the $t$-value shrinks toward the $z$-value.
- By df $\approx 30$ the gap is tiny – the $\infty$ row *is* the normal.

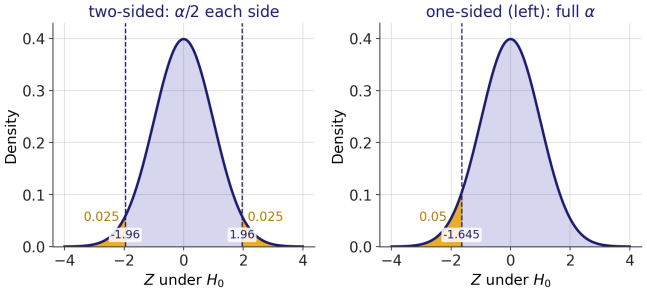
# The wording reveals the test direction

| Signal words | Hypotheses | Direction |
|---|---|---|
| exceed, greater, increased | $H_0 : \mu = \mu_0, \quad H_1 : \mu > \mu_0$ | right-sided |
| fall below, smaller, reduced | $H_0 : \mu = \mu_0, \quad H_1 : \mu < \mu_0$ | left-sided |

**Note**

The *question* decides one- or two-sided – not the data. The statement you want to find evidence for goes into $H_1$.

# Why put the whole $\alpha$ on one side?




- Two-sided spends $\alpha/2$ guarding *each* tail – guarding the harmless side is wasted.
- One-sided puts the *whole* $\alpha$ in the relevant tail, so the critical value moves inward ($1.96 \to 1.645$).
- A true deviation in that direction now clears the bar *more easily*.

## Key takeaway

Same $\alpha$, more power – but *only* in the chosen direction. A one-sided test at level $\alpha$ uses the critical value of a two-sided test at level $2\alpha$ on that side.

# The price: you must commit in advance

**The direction is part of the hypothesis**

Choose one- vs. two-sided **before** looking at the data – based on the research question alone.

- Picking the side *after* seeing which way the data point inflates the true Type I error from $\alpha$ to as much as $2\alpha$.
- A one-sided test is **blind** to deviations in the other direction: even a large effect on the "wrong" side never rejects $H_0$.
- If both directions genuinely matter, use a two-sided test.

**In words:** One-sided tests buy power with a promise – that the other direction really does not matter.

# Left-sided test – consumer protection

## Task

A new filling line fills milk (label 1 l). Are the consumer protection requirements met, i.e. is *not too little* being filled? Test with $1 - \alpha = 0.95$.

- $H_0 : \mu = 1000$ vs. $H_1 : \mu < 1000$.
- Only $\mu < 1000$ (too little in the bottle) is critical.

## Procedure

Test statistic $T_n = \dfrac{\bar{X} - \mu_0}{s/\sqrt{n}}$; reject if $T_n < -t_{1-\alpha,\, n-1}$ (known: $Z_n < -z_{1-\alpha}$).
$p$-value: $p = F_{t,n-1}(T_n)$.

**In words:** We reject only if $\bar{X}$ falls *clearly below* $\mu_0$ – a value above $\mu_0$ can never be evidence for "too little".

# Right-sided test – do not exceed a limit

## Task

Advertisement: "Our milk contains **at most** $1\,\text{mg} = 1000\,\mu\text{g}$ of a contaminant per litre." Is this met? We measure in $\mu$g, so $\mu_0 = 1000\,\mu\text{g}$.

- $H_0 : \mu = 1000\,\mu\text{g}$ vs. $H_1 : \mu > 1000\,\mu\text{g}$.
- *Exceeding* the limit is critical.

## Procedure

Reject if $T_n > t_{1-\alpha,\,n-1}$ (known: $Z_n > z_{1-\alpha}$).
$p$-value: $p = 1 - F_{t,n-1}(T_n)$.

**In words:** Mirror image of the left-sided case: we reject only for values *clearly above* $\mu_0$.

# Right-sided test – substance K with numbers

**Data (known variance):** $\sigma = 90\,\mu$g, $n = 9$, $\bar{x} = 1060\,\mu$g, $\mu_0 = 1000$, $\alpha = 0.05$, $z_{0.95} = 1.645$.

$$Z_n = \frac{1060 - 1000}{90/\sqrt{9}} = \frac{60}{30} = 2.00$$

$$Z_n = 2.00 \;>\; 1.645 = z_{0.95}$$

## Decision

$H_0 : \mu = 1000$ is **rejected** ($p = 1 - \Phi(2.00) = 0.0228$). Evidence that the limit is *exceeded* – the advertising claim is not statistically tenable.

# Left vs. right-sided – at a glance

| | left-sided | right-sided |
|---|---|---|
| Question | "too *small*?" | "too *large*?" |
| $H_0$ / $H_1$ | $\mu = \mu_0$ / $\mu < \mu_0$ | $\mu = \mu_0$ / $\mu > \mu_0$ |
| Reject if | $T_n < -t_{1-\alpha,n-1}$ | $T_n > t_{1-\alpha,n-1}$ |
| Rejection region | far *left* tail | far *right* tail |
| $p$-value | $F_{t,n-1}(T_n)$ | $1 - F_{t,n-1}(T_n)$ |

Some textbooks write the null as $\mu \geq \mu_0$ resp. $\mu \leq \mu_0$ – the test is **identical**, because critical value and $p$-value are computed at the boundary $\mu = \mu_0$ either way.

## Key takeaway

$H_1$ always points *toward* the rejection region: the inequality in $H_1$ tells you which tail you reject in.

# One-sided $p$-value – formulas

## Left-sided

$$p = F_{t,n-1}(T_n)$$

known: $p = \Phi(Z_n)$

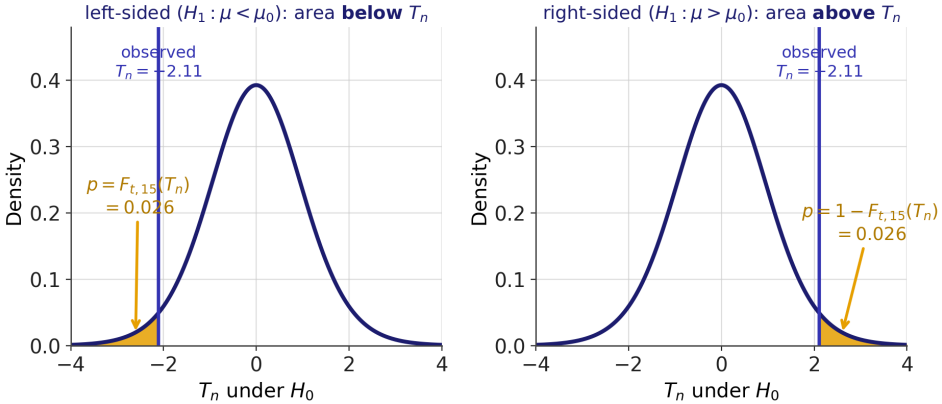
## Right-sided

$$p = 1 - F_{t,n-1}(T_n)$$

known: $p = 1 - \Phi(Z_n)$

**In words:** In a one-sided test we do *not* double – only the one relevant tail area counts.

# One-sided $p$-value – graphically




The $p$-value is the *single* tail area beyond the observed statistic, **in the direction of** $H_1$. Left panel: the cream example ($T_n = -2.11$, df $= 15$, $p = 0.026$).

# One-sided $p$-value – mind the sign

## Only the $H_1$-tail counts

The one-sided $p$-value is the probability, under $H_0$, of a value *at least as far in the $H_1$ direction* as the one observed. The opposite tail is irrelevant – a deviation there supports $H_0$.

## Sign check – a big $|z|$ in the wrong direction is *not* significant

Right-sided test with $z = -2.5$: the statistic is on the $H_0$ side, so $p = 1 - \Phi(-2.5) = 0.994 > 0.5$ – you **never** reject, even though $|z|$ is large. Check the data point in the $H_1$ direction *before* computing.

**In words:** Link to two-sided: if the deviation is in the $H_1$ direction, $p_{\text{one}} = p_{\text{two}}/2$; otherwise $p_{\text{one}} = 1 - p_{\text{two}}/2$.

# One-sided $p$-value – milk ($T_n = 0.7947$, df $= 8$)

## left-sided

$$p = F_{t,8}(0.7947) = 0.7751$$

$> 0.05 \Rightarrow H_0 : \mu = 1000$ kept.

## right-sided

$$p = 1 - 0.7751 = 0.2249$$

$> 0.05 \Rightarrow H_0 : \mu = 1000$ kept.

## Interpretation (left)

No evidence that the machine fills *too little* – from a statistical point of view, consumer protection is met.

# One-sided $p$-value – cream (left, unknown var.)

With $T_n = -2.1086$, $\mathrm{df} = 15$:

$$p = F_{t,15}(-2.1086) = 0.0261$$

## Decision

$p = 0.0261 < 0.05 \Rightarrow H_0 : \mu = 250$ is **rejected** (in favour of $H_1 : \mu < 250$).

## Key takeaway

In contrast to the two-sided test ($p = 0.0522$), the left-sided test is **significant** here – because the whole $\alpha$ lies on the critical side.

# One-sided test with R

```r
# left-sided (H1: mu < mu0)
t.test(milk, mu = 1000, alternative = "less")
# right-sided (H1: mu > mu0)
t.test(milk, mu = 1000, alternative = "greater")
```

## Cream, left-sided – excerpt

```
t = -2.1086, df = 15, p-value = 0.02606
95 percent confidence interval:  -Inf  249.82
```

The argument `alternative` controls the direction (`"two.sided"`, `"less"`, `"greater"`).

# One-sided confidence intervals

- A one-sided test has a one-sided CI – only *one* bound, the other is $\pm\infty$.
- R reports them as $[-\infty;\, x]$ resp. $[x;\, +\infty]$.

## Calculation (known variance)

$$\text{left-sided test } (H_1 : \mu < \mu_0): \quad \left(-\infty,\ \bar{X} + z_{1-\alpha}\frac{\sigma}{\sqrt{n}}\right]$$

$$\text{right-sided test } (H_1 : \mu > \mu_0): \quad \left[\bar{X} - z_{1-\alpha}\frac{\sigma}{\sqrt{n}},\ +\infty\right)$$

For unknown variance: replace $z_{1-\alpha}$ by $t_{1-\alpha,\, n-1}$.

Note the one-sided quantile $z_{1-\alpha}$ (not $z_{1-\alpha/2}$) – the whole $\alpha$ sits on one side, exactly as in the test.

# One-sided CIs – milk and cream (known)

**CI for the *left-sided* test: upper bound** ($z_{0.95} = 1.645$):

$$\text{Milk} : 1001 + 1.645 \cdot \frac{3}{\sqrt{9}} = 1002.645$$
$$\text{Cream} : 248.9375 + 1.645 \cdot \frac{2}{\sqrt{16}} = 249.760$$

**CI for the *right-sided* test: lower bound:**

$$\text{Milk} : 1001 - 1.645 \cdot \frac{3}{\sqrt{9}} = 999.355$$
$$\text{Cream} : 248.9375 - 1.645 \cdot \frac{2}{\sqrt{16}} = 248.115$$

**In words:** If $\mu_0$ lies beyond the single bound, $H_0$ is rejected.

# Exercise – net weight (left-sided, unknown $\sigma$)

## Problem

A package must contain *at least* $\mu_0 = 500\,\text{g}$. A sample of $n = 12$ gives $\bar{x} = 496$, $s = 6$. Is there evidence of underfilling? ($\alpha = 0.05$.)

**Hint:** $t_{0.95;\,11} = 1.7959$.

# Solution – net weight

Underfilling is critical $\Rightarrow H_0 : \mu = 500$ vs. $H_1 : \mu < 500$ (left-sided). Reject if $T_n < -t_{0.95;\,11} = -1.7959$.

$$T_n = \frac{496 - 500}{6/\sqrt{12}} = \frac{-4}{1.7321} = -2.3094 < -1.7959$$

**Decision**

Reject $H_0$: evidence of underfilling. $p = F_{t,11}(-2.3094) = 0.0207 < 0.05$.

# Exercise – one- vs. two-sided on the same data

## Problem

A coating must *exceed* a target thickness. A test gives $T_n = 1.80$ with df $= 15$. Compare the right-sided and the two-sided decision at $\alpha = 0.05$.

**Hint:** $1 - F_{t,15}(1.80) = 0.0460$.

# Solution – one- vs. two-sided

$$\text{right-sided: } p = 1 - F_{t,15}(1.80) = 0.0460 < 0.05 \;\Rightarrow\; \text{reject}$$
$$\text{two-sided: } p = 2 \cdot 0.0460 = 0.0920 > 0.05 \;\Rightarrow\; \text{keep}$$

**Key takeaway**

Identical data, opposite decisions. The one-sided test is more powerful *in the chosen direction* – exactly why the direction must be fixed before seeing the data.

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2 Hypothesis Tests

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4 One-Sided Tests

5 Types of Error
- The two types of error
- $\alpha$, $\beta$ and power
- Exercises

6 Summary

# Two ways to be wrong

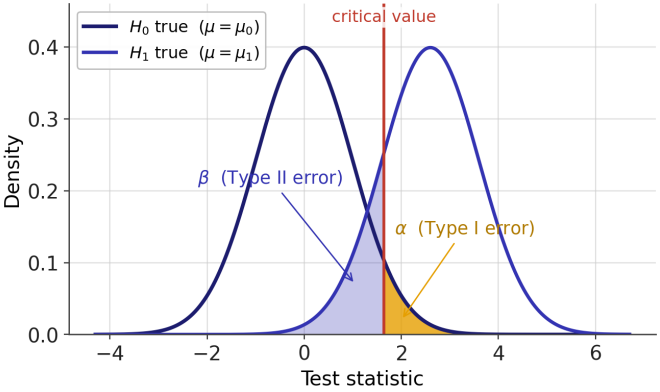
## Intuition

Like a smoke detector, we can be wrong in two ways: **false alarm** or **missed fire**.

| **Decision** | $H_0$ **true** | $H_0$ **false** |
|---|---|---|
| reject $H_0$ | Type I error ($\alpha$) | correct (power $1-\beta$) |
| keep $H_0$ | correct ($1-\alpha$) | Type II error ($\beta$) |

**Type I error** = false alarm (reject although $H_0$ is true). **Type II error** = miss (keep although $H_1$ is true).

# $\alpha$ and $\beta$ in a picture




- The two curves are $\bar{X}$ in each "world": left centred at $\mu_0$ (if $H_0$ holds), right at the true $\mu_1$ (if $H_1$); the red line is the critical value.
- $\alpha$: $H_0$ true, but beyond the critical value.
- $\beta$: $H_1$ true, but on this side.
- The distributions overlap – both errors cannot be 0 at once.

# Type I error ($\alpha$) – controllable

## Definition

$$P(\text{reject } H_0 \mid H_0 \text{ true}) = \alpha$$

- **Directly controllable** through the choice of $\alpha$ (typically 0.05).
- In 5 % of cases a *true* $H_0$ is wrongly rejected.

## Milk

$H_0 : \mu = 1000$ is true, yet the test rejects (false alarm) – probability $= \alpha$.

# Type II error ($\beta$) – only indirectly influenced

## Definition (at a specific alternative $\mu_1$)

$$P(\text{keep } H_0 \mid \mu = \mu_1) = \beta$$

- No *direct* lever; $\beta$ depends on:
  - $\alpha$, sample size $n$, spread $\sigma^2$,
  - true deviation $|\mu - \mu_0|$ (effect size).
- At a fixed $\mu_1$: smaller $\alpha \Rightarrow$ higher bar $\Rightarrow$ more misses ($\beta$ *rises*); larger $n \Rightarrow$ smaller SE $\Rightarrow$ $\beta$ *falls*.

## Key takeaway

Power $= 1 - \beta$: the ability to actually reject a *false* $H_0$.

# Only one error is controllable

**Question:** Which error can we only make when we *reject* $H_0$?

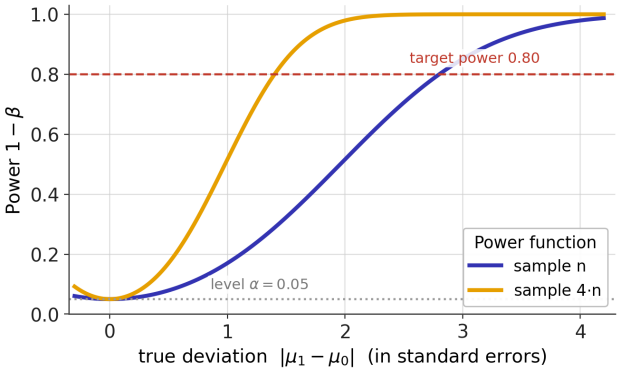
## Answer

The Type I error – and that is exactly the one we control via $\alpha$.

## Consequence for the choice of hypotheses

This is why the "worse" alternative is formulated as $H_1$: the feared error becomes the controllable Type I error.

# Trade-off and power




- $\alpha\downarrow \Rightarrow \beta\uparrow$ (trade-off).
- Larger $n$ raises power at every $\mu \neq \mu_0$ (at $\mu_0$ it stays at $\alpha$) – more power at the same level.
- Power rises with the true deviation.

## Practice

Typically $\alpha = 0.05$ and desired power $1 - \beta \geq 0.80$. A **power analysis** provides the required $n$.

# Computing $\beta$ concretely – milk

**Setup:** $H_0 : \mu = 1000$ vs. $H_1 : \mu \neq 1000$, $\sigma = 3$, $n = 9$ $(\mathrm{SE} = 1)$, $\alpha = 0.05$.

**Step 1** – turn "$|Z_n| \leq 1.96$" into bounds on $\bar{X}$:
$\mu_0 \pm 1.96 \cdot \mathrm{SE} = 1000 \pm 1.96 = [998.04;\ 1001.96]$.

**Step 2** – the truth is $\mu_1 = 1002$; find how often $\bar{X}$ still lands in that region, *standardizing with* $\mu_1$:

$$\begin{aligned}\beta &= P\big(998.04 \leq \bar{X} \leq 1001.96 \;\big|\; \mu = 1002\big) \\ &= \Phi\Big(\tfrac{1001.96-1002}{1}\Big) - \Phi\Big(\tfrac{998.04-1002}{1}\Big) = 0.484\end{aligned}$$

**Result**

Power $= 1 - \beta = 0.516$: a true deviation of 2 ml is detected with $n = 9$ in only just over half the cases.

# Four levers for more power

More sample size helps decisively. Same situation with $n = 36$ instead of 9:

$$\text{SE} = \frac{3}{\sqrt{36}} = 0.5 \;\Rightarrow\; \beta = 0.021, \quad \text{power} = 0.979.$$

## Levers of power $1 - \beta$

1. **Sample size** $n \uparrow$ (strongest lever).
2. **Effect size** $|\mu_1 - \mu_0|$ (given).
3. **Spread** $\sigma \downarrow$ (e.g. measure more precisely).
4. **Level** $\alpha \uparrow$ (buys more Type I error).

## Key takeaway

Quadrupling $n$ from 9 to 36 raises the power from 0.52 to 0.98.

# Exercise – compute $\beta$ and power (two-sided)

## Problem

Test $H_0 : \mu = 100$ vs. $H_1 : \mu \neq 100$, known $\sigma = 10$, $n = 25$, $\alpha = 0.05$. The true mean is actually $\mu_1 = 105$. Find the Type II error $\beta$ and the power.

**Step 1:** the acceptance region for $\bar{X}$ under $H_0$.

# Solution – $\beta$ and power (two-sided)

SE $= 10/\sqrt{25} = 2$. Acceptance region: $100 \pm 1.96 \cdot 2 = [96.08\,;\,103.92]$.

$$\begin{aligned}\beta &= P(96.08 \le \bar{X} \le 103.92 \mid \mu = 105) \\ &= \Phi\Big(\tfrac{103.92-105}{2}\Big) - \Phi\Big(\tfrac{96.08-105}{2}\Big) = \Phi(-0.54) - \Phi(-4.46) \\ &= 0.2946 - 0.0000 = 0.2946\end{aligned}$$

**Power**

$1 - \beta = 0.7054$: a true shift of 5 units is detected in about 71 % of samples.

# Exercise – power of a one-sided test

## Problem

Now use a *right-sided* test, $H_1 : \mu > 100$, with $\sigma = 10$, $n = 25$, $\alpha = 0.05$, true $\mu_1 = 105$. Find the power and compare with the two-sided value 0.7054.

# Solution – one-sided power

SE $= 2$. Critical mean: $\bar{x}_{\text{crit}} = 100 + 1.645 \cdot 2 = 103.29$.

$$1 - \beta = P(\bar{X} > 103.29 \mid \mu = 105) = 1 - \Phi\left(\tfrac{103.29-105}{2}\right) = 1 - \Phi(-0.855) = 0.8038$$

## Comparison

Power 0.804 vs. 0.705 two-sided – the one-sided test is *more powerful* for the same $n$ and $\alpha$, because the whole $\alpha$ sits on the relevant side.

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# Summary 1 – confidence intervals

## Known variance $\sigma$ $(Z_n \sim N(0,1))$

$$\mathrm{CI}_{1-\alpha} = \left[\bar{X} - z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}},\ \bar{X} + z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}\right]$$

## Unknown variance $(T_n \sim t_{n-1})$

$$\mathrm{CI}_{1-\alpha} = \left[\bar{X} - t_{1-\alpha/2,\,n-1}\frac{s}{\sqrt{n}},\ \bar{X} + t_{1-\alpha/2,\,n-1}\frac{s}{\sqrt{n}}\right]$$

# Summary 2 – hypothesis test (two-sided)

$H_0 : \mu = \mu_0$ vs. $H_1 : \mu \neq \mu_0$

## Test statistics

$$Z_n = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}, \qquad T_n = \frac{\bar{X} - \mu_0}{s/\sqrt{n}} \sim t_{n-1}$$

## Decision

Reject if $|Z_n| > z_{1-\alpha/2}$ resp. $|T_n| > t_{1-\alpha/2,n-1}$.

# Summary 3 – the $p$-value

## One idea behind every formula

$p$ = probability, *if* $H_0$ *were true*, of a value **at least as extreme as observed** – in the direction(s) that $H_1$ allows.

| **If $H_1$ is...** | **add up the...** | $p$ **(unknown $\sigma$)** |
|---|---|---|
| $\mu \neq \mu_0$ | *both* tails beyond $\lvert T_n \rvert$ | $2\,F_{t,n-1}(-\lvert T_n \rvert)$ |
| $\mu < \mu_0$ (left) | *lower* tail below $T_n$ | $F_{t,n-1}(T_n)$ |
| $\mu > \mu_0$ (right) | *upper* tail above $T_n$ | $1 - F_{t,n-1}(T_n)$ |

**Known $\sigma$?** Just replace $T_n \to Z_n$ and $F_{t,n-1} \to \Phi$. **Decide:** $p < \alpha \Rightarrow$ reject $H_0$.

# Summary 4 – one-sided tests & types of error

## One-sided tests (unknown variance)

**left:** $H_1 : \mu < \mu_0$ reject if $T_n < -t_{1-\alpha,\, n-1}$

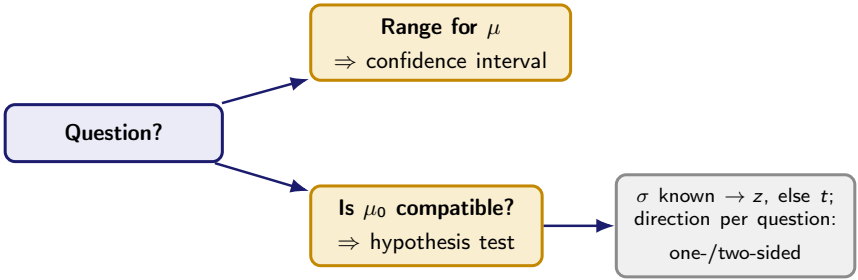
**right:** $H_1 : \mu > \mu_0$ reject if $T_n > t_{1-\alpha,\, n-1}$

## Types of error

$P(\text{reject } H_0 \mid \mu = \mu_0) = \alpha$ $P(\text{keep } H_0 \mid \mu = \mu_1) = \beta$

$\alpha$ is *controllable*; $\beta$ falls with $n \uparrow$ and $\alpha \uparrow$; power $= 1 - \beta$.

# Which procedure? – decision tree




The same logic – *standardize*, then ask whether the deviation is extreme *in the $H_1$ direction* – underlies essentially every classical test: $z$, $t$, proportion, regression coefficient, ANOVA $F$.

## Key takeaway

First clarify the *question*, then choose the tool.

# Five classic pitfalls – final checklist

| **Forbidden sentence** × | **Correct version** ✓ |
|---|---|
| "$p$ is the probability that $H_0$ is true." | $p = P(\text{data at least this extreme} \mid H_0)$. |
| "$\mu$ lies in *this* CI with 95 % probability." | The *procedure* covers $\mu$ in 95 % of samples. |
| "$p > \alpha$, so $H_0$ is proven/accepted." | No contradiction found – $H_0$ is *kept*, not proven. |
| "The data point left, so I test left-sided." | Direction and $\alpha$ are fixed *before* the data. |
| "Significant, hence large and important." | Significance $\neq$ size: judge magnitude by the CI. |

# Outlook: Part 4 – Regression

## What we have so far

Confidence intervals and tests for **one** population mean $\mu$: estimate, standard error, critical value – and the $p$-value.

## Next part

We study the **relationship between variables**. The regression slope $\hat{\beta}_1$ is again an estimator with a standard error, so the *same* recipe gives its $t$-test, its confidence interval and its $p$-value:

$$T = \frac{\text{estimate} - \text{hypothesized value}}{\text{standard error}}.$$

## Key takeaway

Everything in this part carries over one-to-one to the coefficients of a regression model.